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Education

- **Harvard College** — A.B. Computer Science
  3.8 GPA / ACT 35, John Harvard Scholar (2023-2024) — *Expected May 2027*
  - Coursework: Distributed Systems (CS2620), Operating Systems (CS1610), Modern Data Storage Systems (CS2640), Computer Architecture (CS1411), Machine Learning, Algorithms, Probability, Linear Algebra

Work Experience

- **Ironsite** — San Francisco, CA
  Member of Technical Staff — Infrastructure Engineer — *May 2026 - Present*
  - Contribute to Ironsite's ML research orchestrator (Rust + Postgres): shipped config-driven inference-engine selection between SGLang and vLLM with build-time engine resolution and dispatch across two GPU image variants.
  - Own the Ansible + OpenTofu infrastructure for a GPU Docker Swarm cluster: Docker + NVIDIA Container Toolkit install, Tailscale networking, and AWS-side EC2 / IAM / S3 / Secrets Manager resources.
  - Designed the cluster's observability stack: a single Vector agent per node scrapes `node_exporter` and `dcgm-exporter` GPU metrics to VictoriaMetrics and ships container logs to VictoriaLogs.
  - Partner with research engineers to translate training and inference requirements into platform capabilities, including an internal-apps platform (EC2 + Traefik) that ships new dashboards on a single-manifest merge.
- **Harvard Generative AI Research Program** — Cambridge, MA
  Research Fellow — *June 2025 - Aug 2025*
  - Designed distributed deep learning pipelines executing across GPU clusters using SLURM orchestration and dynamic resource allocation.
  - Integrated Bayesian optimization into existing OpenFold workflows, achieving 2× validation improvement while maintaining scalable execution across compute nodes.
- **Los Alamos National Laboratory, B-GEN** — Santa Fe, NM
  Student Researcher — *June 2022 - Aug 2023, Summer 2024*
  - Optimized distributed HPC pipelines processing terabyte-scale biological datasets, reducing workflow runtime by 80% through SLURM parallelization and memory tuning.
  - Engineered GPU-accelerated data processing modules for large-scale biological computation, eliminating throughput bottlenecks in production research workflows.

Selected Systems & Performance Projects

- **RCache-RDMA: One-Sided RDMA Distributed Cache (C++)** — Harvard CS2640
  Networking and Distributed Systems Project — *2026*
  - Built a key-value cache with three communication paths in C++ using `libibverbs`: TCP/RPC baseline, two-sided RDMA messaging, and one-sided RDMA reads over a registered hash-table memory region.
  - Achieved 974k ops/s with stable 12 μs p99 latency on one-sided RDMA — a 13× throughput improvement and 20× tail-latency reduction over the TCP baseline on CloudLab Mellanox hardware.
- **Adaptive Stream Buffer Prefetcher (C++, Intel Pin)** — Harvard CS1411
  Computer Architecture Project — *2026*
  - Implemented Jouppi's fixed-depth stream buffer and Palacharla-Kessler's adaptive prefetch policy as Intel Pin tools, modeling a two-level cache hierarchy with explicit latency costs; a histogram-driven depth selector learned stream-length distributions online, achieving 5.02× speedup on libquantum and 83% prefetch accuracy on dealII.
- **High-Performance Async RPC Client (C++, gRPC)** — Harvard CS2620
  Distributed Systems Project — *2026*
  - Designed an asynchronous RPC client using gRPC completion queues and a multi-threaded polling architecture.
  - Increased throughput from ∼8k to 55k+ RPC/s via batching, non-blocking I/O, and flow-control tuning.
- **Chickadee OS Kernel (C++)** — Harvard CS1610
  Operating Systems Project — *2026*
  - Implemented blocking system calls (`sys_waitpid`, `sys_msleep`) with a timer-driven sleep mechanism backed by a hashed timing structure — eliminating busy-wait loops while maintaining correctness under multi-core execution.

Skills

- **Languages:** C++, Python, Rust, Bash
- **Systems:** Linux, Concurrency, Distributed Systems, Low-latency Networking, RDMA (`libibverbs`), Intel Pin
- **Infrastructure:** AWS, OpenTofu / Terraform, Ansible, Docker Swarm, Postgres, SLURM, gRPC, Git